In-vitro Biology

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Cell based assays

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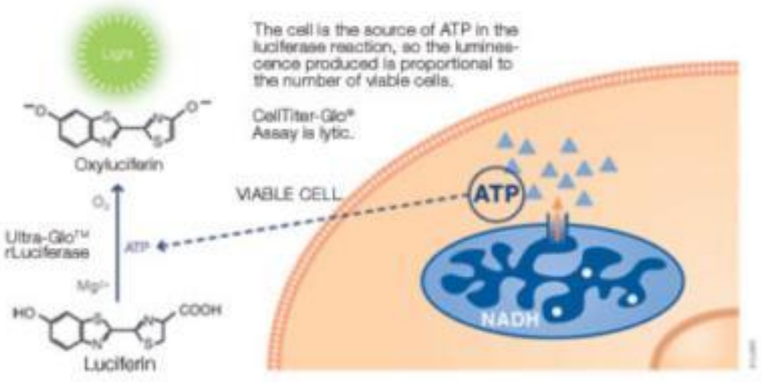
Case study: Cell Proliferation by Cell Titre Glo

Assay principle:



Brief Protocol:

Seed cells in 40µL of growth media with ; 250, 500 & 1000 cells/well.

Overnight, 37°C, 5 % CO_2

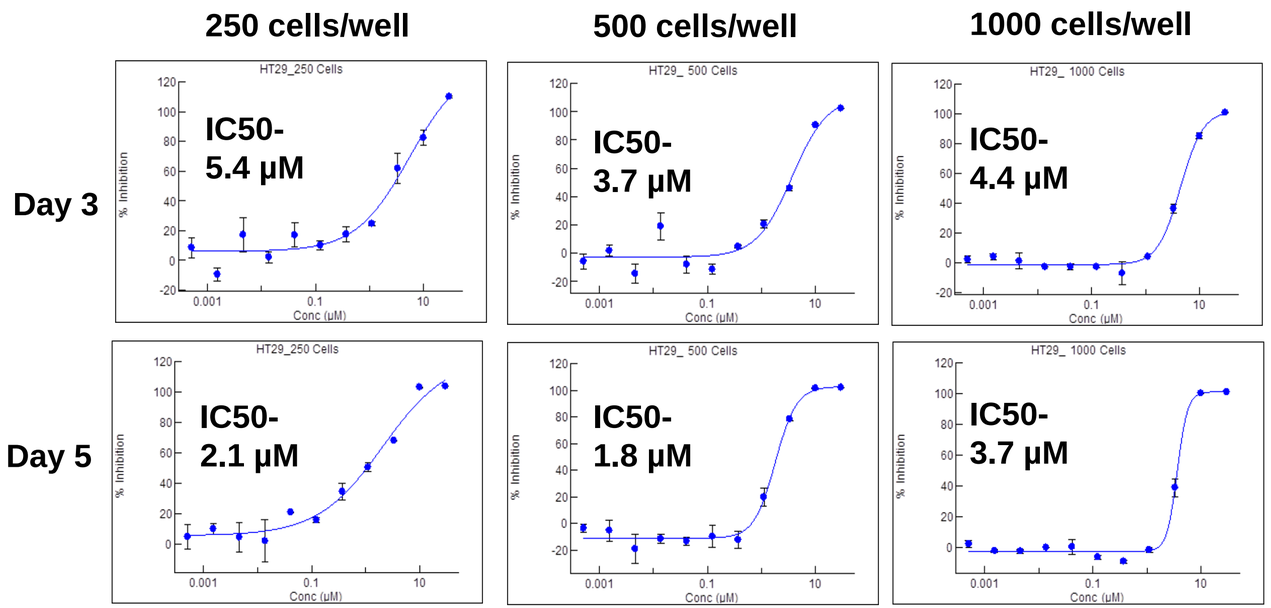
Treat with 10 µL of 5X compound stock. Final DMSO - 0.5%

Incubation for 3, 5 days, 37°C, 5 % CO_2

Termination- Add equal volume of CTG (50 µL), incubate at RT for 10 mins. Read on Envision plate reader

Kit Used- CellTiter-Glo®- *Promega (G7571)*

Cell Number & Treatment Time Optimization in HT29



From the above experiment Day 5 with 500 cells/well were taken forward

Quantification: Luminescence
Instrument used: Envision 2105
QC parameters used for assay:
a) % CV – < 20 %
b) Z' > 0.5
c) Signal Window : 25

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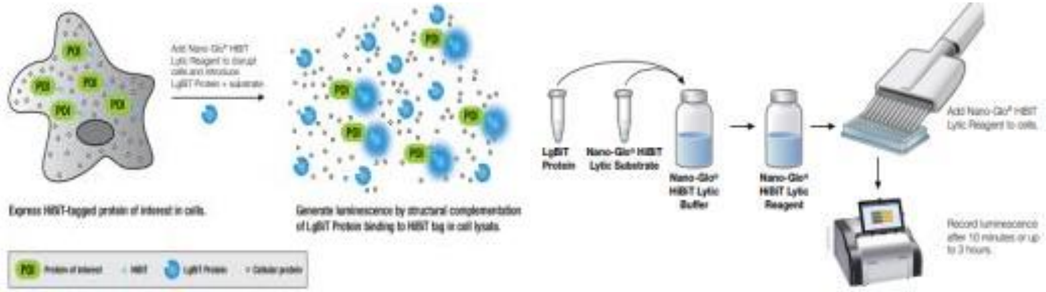
In-vitro Selectivity assay - HiBiT assay

Assay principle:



Brief Protocol:

Seed cells in 30µL of growth media; 1000 cells/well.

Overnight, 37°C, 5 % CO_2

Treat with 10 µL of 4X compound stock.

Incubation for 24 hr, 37°C, 5 % CO_2

Termination- Add 40 µl of HiBiT lytic reagent
incubate at RT for 10 mins. Read luminescence on VictorX5 plate reader

Kit Used- Nano-Glo® HiBiT Lytic Detection System- *Promega (N3040)*

DRC of tool compound in HiBiT cell line

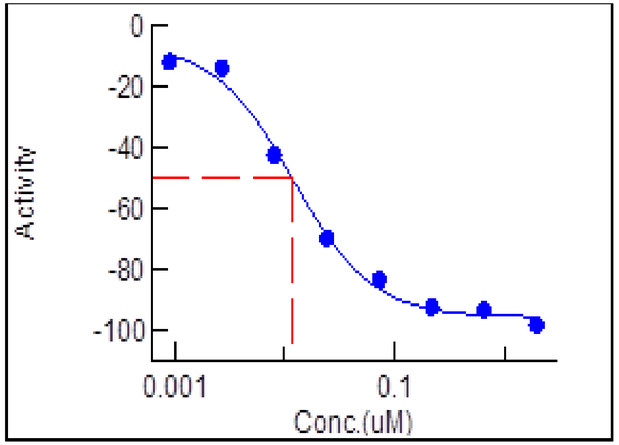


Plate format: 384-well
Quantification: Luminescence
Instrument used: Victor X5 multimode plate reader
QC parameters used for assay:
a) % CV: < 20 %
b) Z': > 0.4
c) Signal Window : 5

~ 1300 compounds screened for an FTE programme using this assay technology. HiBiT cell line was provided by client..

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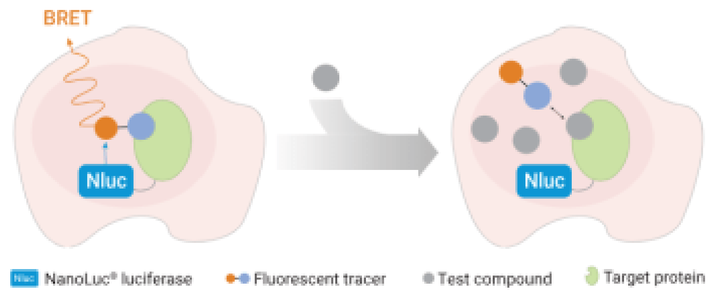
NanoBRET Target Engagement Intracellular kinase Assay

Assay principle:



Brief Protocol:

Day 1- Cell density – 0.2 x 10^6 /ml
Target Vector: Carrier DNA – 1:9
(Target Vector: Carrier DNA) : FuGENE – 1:3
Seed transfected cells in 96 well solid white plate

↓ Overnight, 37°C, 5 % CO_2

Day 2: Add Tracer + Test Compound

↓ 2hrs, 37°C, 5 % CO_2

Add NanoBRET substrate

↓

Read plate in Envision Plate Reader
Donor Emission – 460 nm; Acceptor Emission – 610 nm
Data representation - NanoBRET Ratio (Acceptor/Donor)

Kit Used- NanoBRET™ TE Intracellular Assay - *Promega (N2630)*

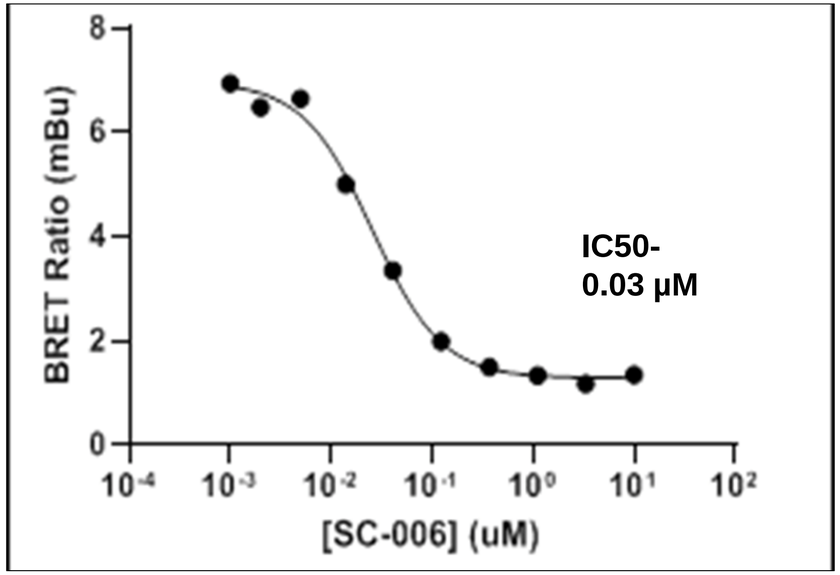


Plate format: 96 well
Quantification: Luminescence & Fluorescence
Instrument used: Envision 2105
QC parameters used for assay:
a) % CV: < 20 %
b) Z': > 0.5
c) Signal Window : 5

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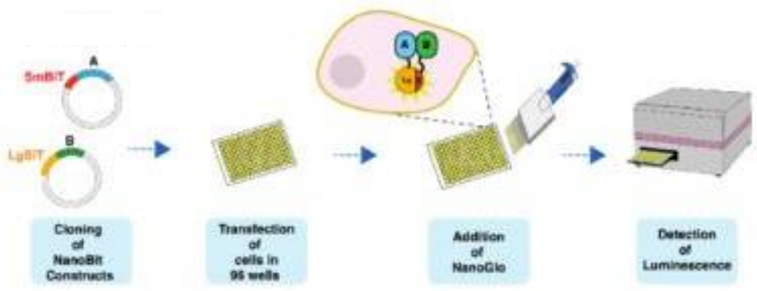
NanoBiT Assay

Assay principle:



Brief Protocol:

smBiT plasmid (10 µg) and lgBiT plasmid (10 µg) were mixed in OptiMEM medium (1000 µL) followed by addition of 60 ul of Fugene HD Transfection.

Incubate at RT for 20 mins

Add HEK293 cells to the transfection mixture at the density of 10 million per 10 mL . Seed 18.5 ul per well in 384 white plate.

Overnight incubation37°C, 5 % CO_2

Add 2.5 ul of neddylation inhibitor (MLN4924)

Overnight incubation37°C, 5 % CO_2

Add 3 µl of test compound and 6 µL of NanoBiT substrate and measure luminescence in Envision plate reader after 30 mins

Kit Used- Nanoglo Live Assay System- ***Promega (N2012)***

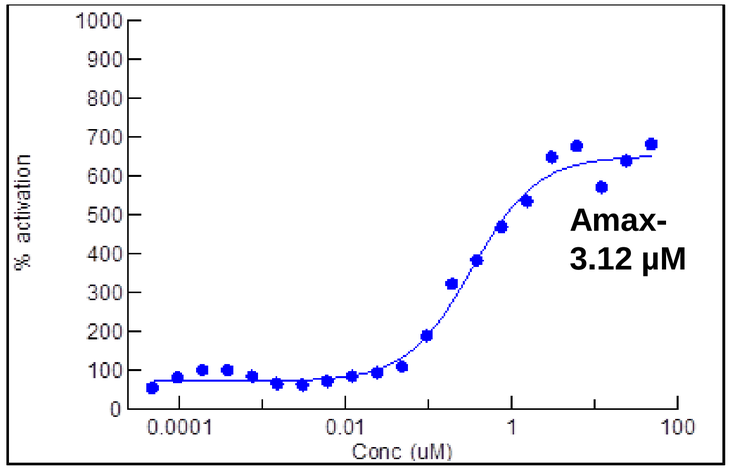


Plate format: 96 well
Quantification: Luminescence
Instrument used: Envision 2105
QC parameters used for assay:
a) % CV: < 20 %
b) Z': > 0.5
c) Signal Window : 5

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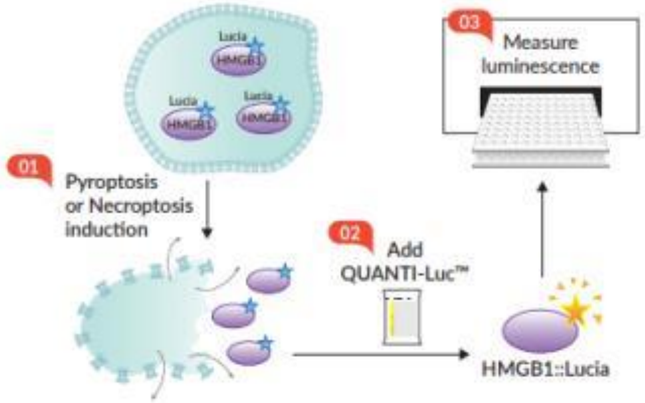
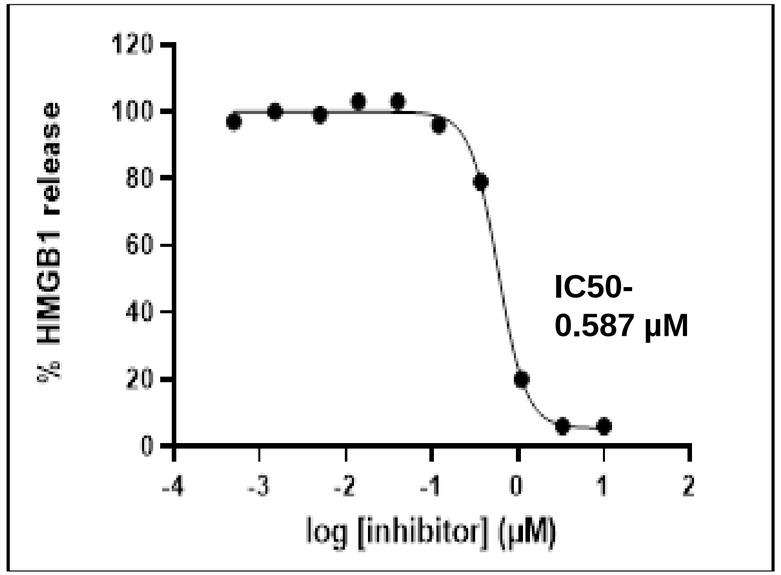
Necroptosis Assay (HMGB1 Release)

Assay principle:





Brief Protocol:

Seed 6 X 10^4 THP1-HMGB1-Lucia cells in 96 well plate

↓

Add pan-caspase inhibitor, 25 µM of Zvad-FMK for 1 hr at 37^0C, 5% CO_2.

↓

Add 100 ng/mL of TNFα, 5 µM of BV-6 and inhibitor compounds

↓ 16h incubation, 37°C, 5 % CO_2

Collect 30 µl of cell culture supernatant in 96 well Opti-plate

↓

Add 150 µl Quanti-Luc substrate, mix and measure luminescence in Envision plate reader

Plate format: 96 well
Quantification: Luminescence
Instrument used: Envision 2105
QC parameters used for assay:
a) % CV: < 20 %
b) Z': > 0.5
c) Signal Window : 5

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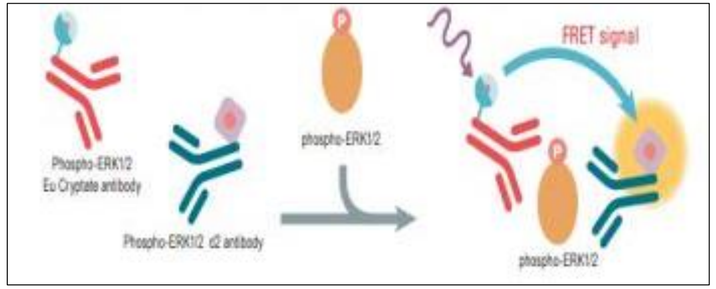
pERK HTRF of Compound X in AGS & AsPC1 cell line –3 HR treatment

Assay principle:



Brief Protocol:

Day 1: : Seeding of cell lines
384 well plate (8000 cells/well)

↓ Overnight, 37°C, 5 % CO_2

Day 2: Compound Treatment
(11 points 3 folds – Top Conc – 3 µM)

↓ Incubate for 3 hrs

Add Lysis Buffer 8 µL of 4X stock

↓ Incubate for 45 mins

Transfer 16 ul of sample to proxiplate. Add 4 ul of antibody mix
(pERK Eu cryptate + pERK d2 antibody)

↓ Incubate for 24 hrs at RT

Reading at EX-340 nm; EM – 620 nm, 665 nm

Kit Used- new ERK Advanced kit - *Cisbio (64AERPEG)*

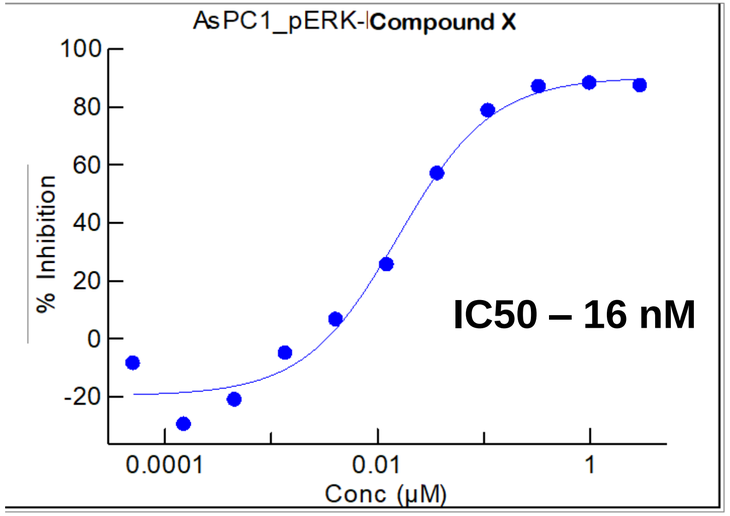


Plate format: 384 well
Quantification: Luminescence
Instrument used: Envision 2105
QC parameters used for assay:
a) % CV: < 20 %
b) Z': > 0.5
c) Signal Window : 5

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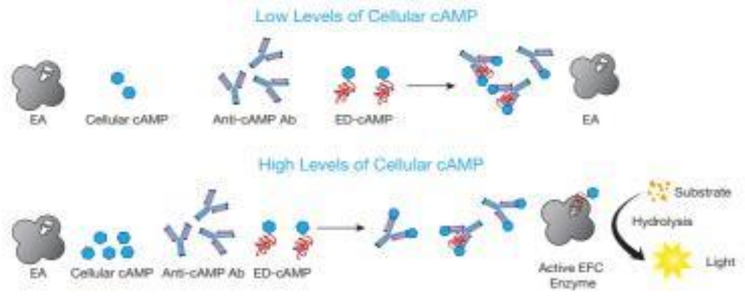
cAMP Hunter™ Liraglutide Bioassay - Protocol

Assay principle:



Brief Protocol:

Day 1: : Cells are thawed and plated in 96 well plate (100 ul/well).

Overnight, 37°C, 5 % CO_2

Day 2: Add cell assay buffer (30 ul) and treat with test compounds (15 ul)

Incubate for 30 mins

Add cAMP antibody reagent and cAMP working detection solution

Incubate for 1hr at RT

Add 60 uL cAMP solution A

Incubate for 3 hrs at RT

Reading chemiluminescence in Envision plate reader

Kit Used- cAMP Hunter™ Liraglutide Bioassay Kit - DiscoverX (95-0062Y2-00099)

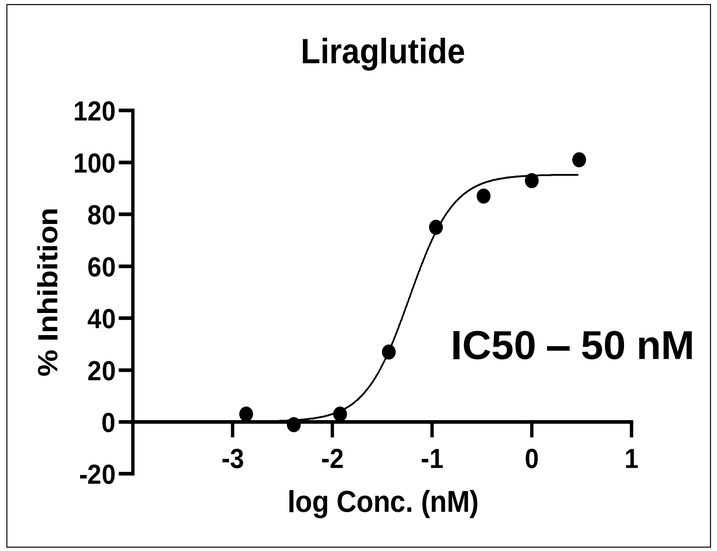


Plate format: 96 well
Quantification: Luminescence
Instrument used: Envision 2105
QC parameters used for assay:
a) % CV: < 20 %
b) Z': > 0.5
c) Signal Window : 5

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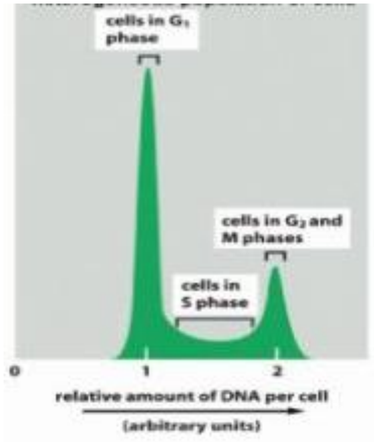
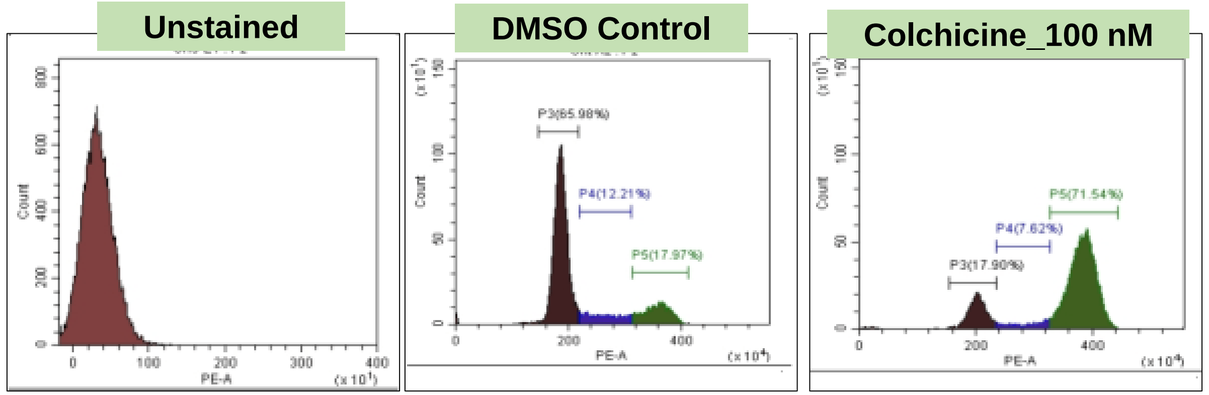
Cell Cycle analysis in the presence of compound X in NCIH358 cell line – 24 hr

Assay principle:

The cell cycle profile of a sample can be determined by staining the DNA with a fluorescent dye and measuring its intensity. The dye stains DNA stoichiometrically, allowing differentiation of cells in G0/G1, S phase, and G2/M.





NCIH358	G0-G1	S	G2-M
Unstained	0.0	0.0	0.0
DMSO Control	66.0	12.2	18.0
Colchicine_100 nM	17.9	7.6	71.5

Brief Protocol:

Day 1: Seeding of cell line- 6 well plate (0.5 * 10^6 cells/well)

Overnight incubation 37C, 5 % CO2

Day 2: Test Compound Treatment

Overnight incubation 37C, 5 % CO2

Day 3: Cells washed with Cell cycle buffer and fixed with 70 % ethanol

Cells washed with Cell Cycle Buffer and stained with Dye Solution for 30 mins at RT

Cells washed and acquired in BD Cytoflex

Kit Used- Cell Cycle – *Sigma (MAK344)*

Plate format: 6 well plate
Quantification:
Instrument used: Cytoflex
QC parameters used for assay:

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Acid accumulation assay

Seed 4 X 10^4 wild type and knockout cells in 96 well plate

↓

Add test compound – 1uM, 5 point DRC with 3-fold dilution

↓

Incubate assay plate for 0, 0.5, 1, 3, 24 & 48h

↓

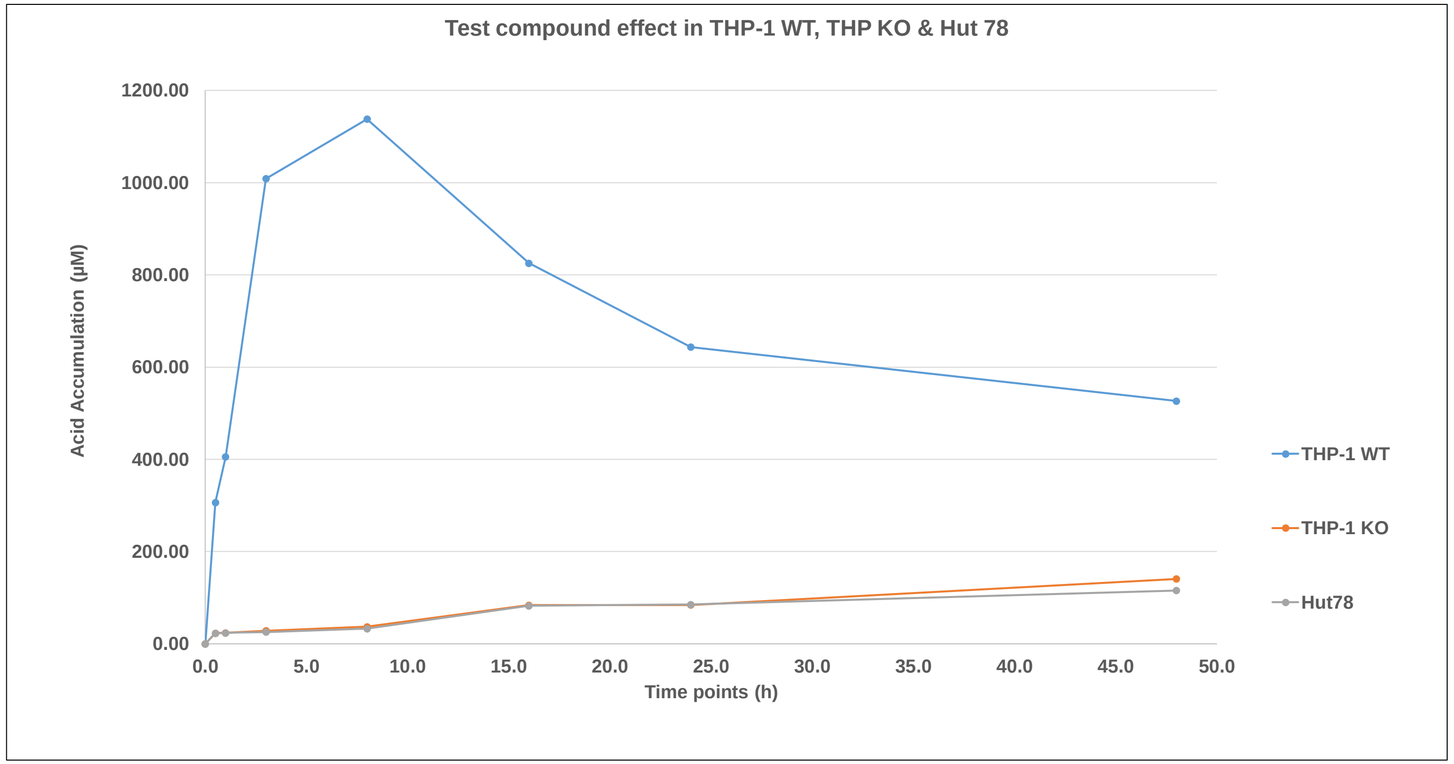
Lyse cells and collect supernatant

↓

Send samples for LCMS analysis

↓

Estimate the acid content compared to ester in the supernatant



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Effect of IFX conjugates on TNFα induced Cytotoxicity in L929 cells

Protocol

Cell Seeding: 2000 and 5000 cells/ well in culture medium (EMEM + 10% HS+ 1% PenStrep)

↓

Incubation for 24h at 37C, 5% CO2

↓

IFX conjugates and TNFα treatment
Assay medium =EMEM+1% HS+ 1%Penstrep
TNFa: Final Concentration used= 5 ng/mL
Concentrations tested: 10000 ng/mL ; 3-fold dilution; 8-point DRC
Treatment time: 24h

↓

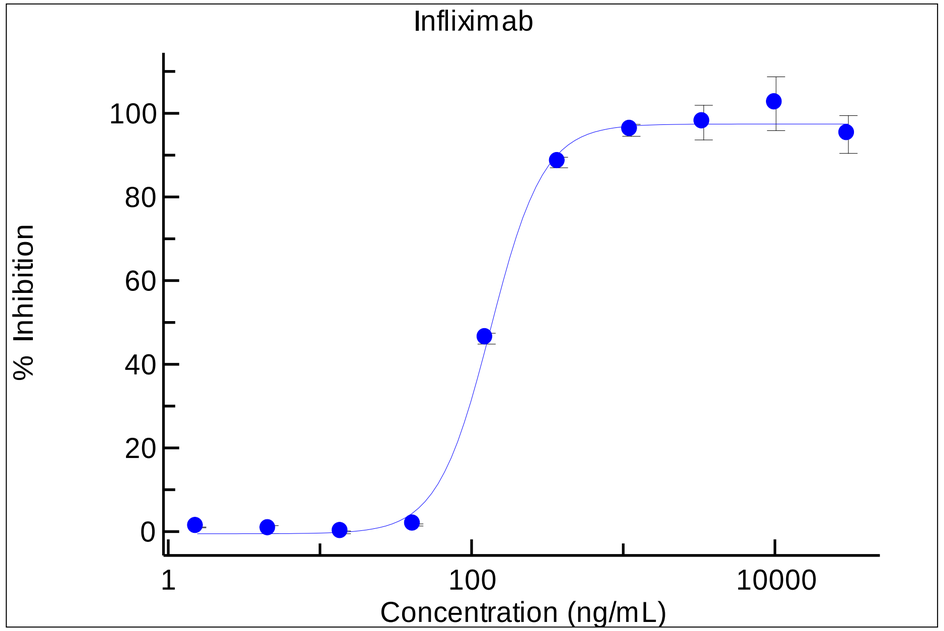
Incubation for 24h at 37C, 5% CO2

↓

Cell Titer Glo Addition

↓

Luminescence Reading



Infliximab			
Fit Cell	#Ok	**IC50 (ng/mL)**	**131.02**
Chart Cell	#Ok	**Hill slope**	2.48

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pSTATs expression in CD4/CD33 cells by Flow cytometry

Whole blood processing

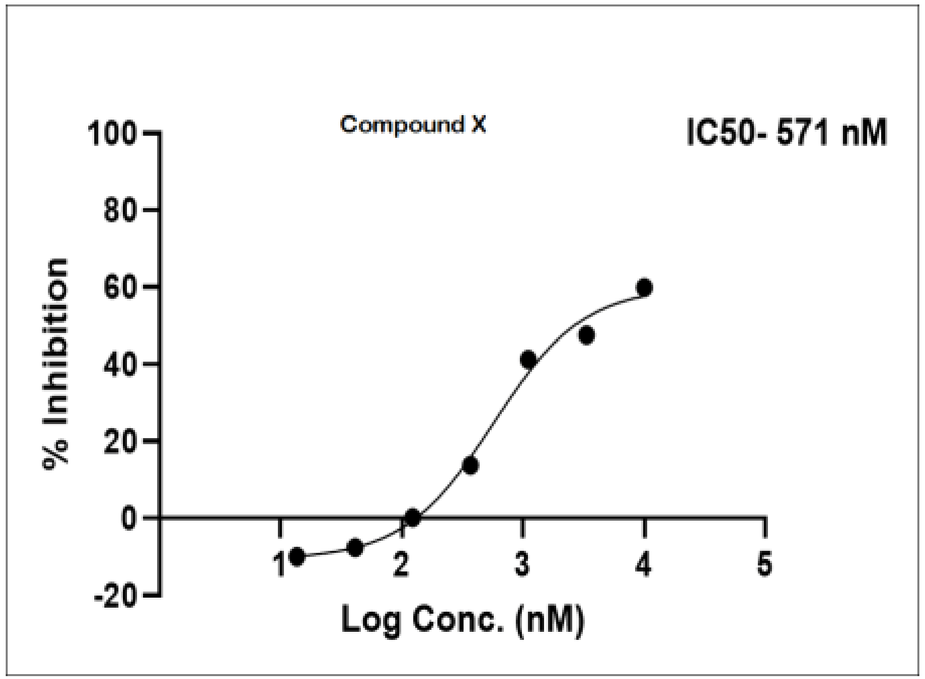
- Blood from healthy donor withdrawn
- Whole blood used for the assay

Compound **treatment** and cytokine induction

- Treatment with compounds - 37°C for 30 min.
- Induction with cytokines - 37°C for 20 min.
- Fixed cells, quick wash

Analysis

- Followed by surface Abs staining at 4°C for 30 min.
- Permeabilization followed by pSTAT staining at 4°C for 2h
- Expression check using flow cytometer



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